

A Comprehensive Study of a Universal Non-Parametric Mesh Magnetic Reluctance Network Model for Radial Interior Permanent Magnet Synchronous Machines

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Abstract—In this paper a universal non-parametric mesh magnetic reluctance network (MRN) model has been proposed for the modelling of radial interior permanent magnet synchronous machines (PMSMs). The main idea of the proposed MRN is to generate the mesh directly by filling the geometry with two-flux-path sector elements automatically in the whole physical region. Then, the circuit topology, excitation vector, and incidence matrix are also formed automatically based on the non-parametric mesh, which are prerequisites for the following circuit analysis solved by nodal analysis. The proposed MRN model can accurately take the nonlinear material property, rotor movement, and period or anti-period boundaries into consideration. Electromagnetic performances and even the magnetic flux density distribution can be solved thoroughly by the proposed MRN model, which are as accurate as the results solved by finite element analysis (FEA). In addition, the non-parametric mesh MRN model is also proven to have powerful parallel computing potentials, and the convergence has been further strengthened by introducing a damping search algorithm and setting proper initial conditions in time-stepping analysis. Finally, the proposed MRN model has been verified by comparisons to both the experimental and FEA results. All these properties of the proposed universal MRN model make it a potential substitution for the FEA model.

Index Terms—Non-parametric mesh, magnetic reluctance network (MRN), electromagnetic modelling, permanent magnet synchronous machine (PMSM), finite element analysis (FEA).

I. INTRODUCTION

PERMANENT magnet synchronous machines (PMSMs) nowadays have become more and more popular than ever before because of the rapid development of electrical vehicles, home appliances, and wind power generators, etc. in recent decades [1], [2], [3]. However, the latest modern topologies of PMSMs bring some extra challenges for the modelling tech-

niques throughout the design stage. This paper mainly focuses on the electromagnetic modelling technique of radial interior PMSMs, in which the high accuracy, fast speed, and easy implementation are the main pursuits.

To achieve fast and accurate electromagnetic modelling of PMSMs, different methods have already been reported in literatures, which can be mainly classified into numerical and analytical techniques [4], [5], [6], [7]. Among all the numerical methods, the finite element analysis (FEA) might be the most popular one because of the accuracy, flexibility, and reliability [8], [9], [10]. However, the main disadvantage is that sometimes FEA consumes too much computational resources and commercial packages might not be affordable for medium and small-sized enterprises, and to develop the FEA software by their own or using open-source packages sometimes may be challenging, e.g., Elmer and ONELAB [11], [12].

The analytical approach would be a good substitution for FEA because of the easy implementation. The magnetic reluctance network (MRN) model might be the most promising approach because a tradeoff can be achieved between the accuracy and efficiency [13], [14]. A. Hemeida et al in [15] proposed a simple and fast MRN model for surface axial flux PMSMs, in which the simulation time was about 600 times faster than the FEA model and the maximum error deviation was about 6% for loss computations. J. Wang and J. Zhu in [16] further extended the usage of the MRN to formulate the eddy current of the permanent magnet and its reaction flux for an axial flux eddy current coupling. Consequently, the maximum torque error was about 11.5% in high slip range. Y. Kano et al in [17] presented an MRN model for a surface-mounted PMSM, in which the nonlinear magnetic material property and the local magnetic saturation were considered. And the simulated errors in comparison to FEA was about 3.5%, while the computation time was reduced from 32 min by FEA to 1.5 s by MRN. G. Gallicchio et al in [18] proposed an accurate hybrid FEA-MRN model for synchronous reluctance machines, in which the nonlinearity was considered by the MRN and the cross-saturation was solved by FEA in the i_d - i_q plane. Consequently, the average torque error solved by the hybrid approach was about 6% compared to the experimental result. Recently, B. Miroschnitschenko in [19] further applied the MRN to model an electromagnetic actuator in 3D domain,

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and the simulation time has been significantly reduced without losing too much accuracy compared to the 3D FEA.

All in all, it can be noticed that the high accuracy and efficiency of the MRN model have already been proven in many literatures and real applications [20], [21], [22], [23], [24], [25], [26], [27] covering different types of electrical machines. However, the main challenge for the MRN model is that it is not an easy task to form a general mesh for the model. In most cases the reluctance map needs to be created manually as studied in [16], [17], [23], [28], which is referred to the parametric mesh MRN model. In these literatures, the mesh for each physical region was generated separately, therefore different and nonuniform basic elements were needed to model different physical regions as reported in [16] and [17]. Consequently, this kind of parametric mesh sometimes needs to be generated manually because of the identifying of the boundary in different physical regions, especially the case small structures are involved. Besides, the irregular flux routes in the air-gap and teeth regions bring an extra challenge to connect the mesh nodes between the stator and rotor as discussed in [28], which further cause difficulties in modelling the rotor movement. To solve this problem, the non-conformal mesh for the MRN has been proposed in [23] to consider the rotor movement by the Lagrange interpolation method on the mesh nodes, therefore remeshing can be avoided despite the increased number of variables due to the interpolation. It can be noticed that the main advantages of the parametric mesh MRN model are easy implementation and quick simulation speed, while the main weakness might lie in the universal property.

To overcome this problem, the non-parametric mesh methodologies by using uniniform mesh elements as reported in [20], [21], [29], [30], [31] are in high demand for the MRN modelling. For the non-parametric mesh model, ideally, a single basic element is needed to create the mesh in the whole region. The physical boundary of each region can be identified by a certain algorithm, and all the pre-processing, solving, and post-processing aspects can be accomplished by programs without manual intervention. Recently, A. M. Silva et al in [30] proposed a general and enhanced MRN model for the analysis of rotating radial-flux machines and the concept of the non-parametric mesh had been introduced and realized. To be more specific, the non-parametric MRN refers to the mesh strategy that the discretization level was independent from the underlying geometry as stated in [18]. However, in [30] the air gap was modelled by a kind of improved air-gap permeance function, which increased the complexity of the modelling. To solve this problem, G. Liu et al. in [20] and D. Cao et al. in [21] proposed new mesh generation methods by using quadrilateral and hexagonal segmentation elements to create the non-parametric mesh MRN model for the air-gap of a linear permanent machine, in which irregular flux distributions between the stator and rotor regions were modelled by the non-parametric mesh so that the rotor movement and air-gap harmonics could be accurately considered. Recently, H. Diab et al in [31] proposed to use triangular elements to generate the mesh for the MRN, which made it possible to model any shape of studied devices and broaden the applicability of the MRN. Nevertheless, the very accurate magnetic field distribution in the

whole region could be hardly obtained, which was important for the prediction of the field related performances (e.g., core losses). It can also be noticed that the main advantages of the non-parametric mesh MRN model are high accuracy, good repeatability, and universal property.

Apart from the mesh generation, the MRN method has also attracted many attentions towards the nonlinear iteration problem during the solving process as discussed in [15], [30], [32], [33]. The relaxation iteration method was used to conduct the nonlinear iteration, and this method might result in difficulties in finding the correct relaxation factor to accelerate the iteration as reported by A. M. Silva et al in [30]. To solve this problem, A. Hemeida et al in [15] used the Newton-Raphson method to accelerate the nonlinear iteration for the MRN model. However, the derivation of the Jacobian matrix was a heavy computational task, and the material's $B-H$ curve had to be modified into a monotonic curve to guarantee the convergence. H. W. Derbas et al in [33] derived the Jacobian matrices for Newton-Raphson iterations for both nodal- and mesh-based MRNs, and it was noticed that the large condition number of the Jacobian matrix of the nodal approach caused difficulties in convergence. B. Boomiraja et al in [32] proposed a simplified Jacobian matrix for the Newton-Raphson method by eliminating the partial derivative term, and the convergence was achieved within a few iterations by using the proposed method. While, difficulties in convergence had been found as well, if the Newton-Raphson method was used for the nodal-based MRN model and the initial permeability was not selected close to the solution.

The aim of this paper is to introduce a universal non-parametric mesh MRN model for radial interior PMSMs covering pre-processing, solving, and post-processing aspects, of which the main advantages are high accuracy, easy implementation, fast convergence, strong robustness, and good repeatability. It is the non-parameter mesh, parallel computing, proposed damping coefficient search algorithm, and detailed comparison between the fixed-point and Newton iterations in this paper, all these together that contribute the novelty of this paper. The innovation of this paper includes the following aspects. The mesh is generated by the non-parametric strategy so that the discretization level is independent from the underlying geometry, and therefore the universal property and automatization are guaranteed. The convergence problem of the proposed non-parametric mesh MRN has been overcome by introducing an extra damping coefficient search algorithm for the fixed-point method. Finally, it shows better applicability, faster speed, more powerful parallel computing capability, and stronger robustness for solving the large-scaled-mesh model by in-depth comparisons with the traditional fixed-point and Newton-Raphson methods. These features further strengthen the practicality and novelty of the proposed MRN method.

This paper is organized as follows: Section II explained the proposed non-parametric mesh MRN model; Section III introduced the corresponding post-processing techniques; Section IV showed the comparisons between the experimental and simulated results. Section V discussed the methods to accelerate the convergence in nonlinear iteration and also the mesh size effects on both accuracy and simulation speed.

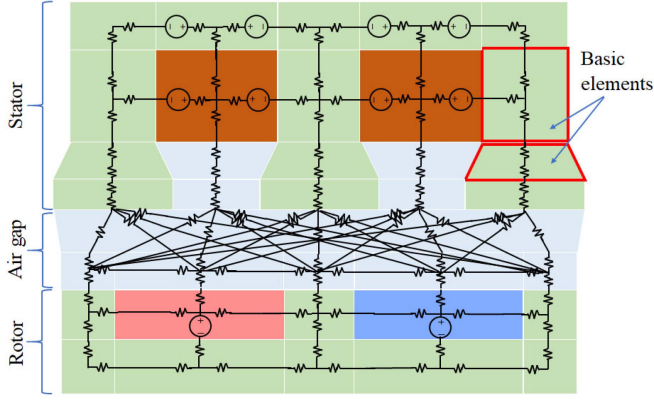


Fig. 1. An example of traditional parametric mesh for a PMSM revised from [30] (for sake of simplicity, a surface-mounted PMSM is modelled).

II. PROPOSED NON-PARAMETRIC MESH MAGNETIC RELUCTANCE NETWORK MODEL

A. Element Description and Non-Parametric Mesh Generation

According to [28], [34], [35], to make it clear to clarify the non-parametric mesh-based MRN model, the parametric mesh strategy is first introduced in this section. Fig. 1 shows an example of traditional parametric mesh for a PMSM, in which two basic elements (rectangle and trapezium) are involved. The procedure of creating the traditional parametric mesh for a radial PMSM can be arranged as follows

- 1) Create 2-D geometry;
- 2) Select several basic elements from rectangle, trapezium, sector, circle, etc.;
- 3) Discretize the geometry segments based on different physical regions such as yoke, tooth, air gap, etc.;
- 4) Fill the physical regions with different basic elements;
- 5) Node and branch numbering;
- 6) Build mesh map and circuit topology.

It can be seen in Fig. 1 that the traditional parametric mesh is relatively coarse (which means that it is difficult to plot the magnetic field distribution) and it involves several basic elements with different shapes. The element size varies a lot with the geometry size and sometimes the very small geometry cannot be modelled accurately because of the lack of the corresponding basic element. In addition, to model the air-gap harmonics accurately, the node connection between the stator and rotor becomes very complex as shown in Fig. 1. And another approach is to use Fourier analysis to model the air gap as reported in [28], which may bring some extra challenges for the following node and branch numbering. It can be noticed that the whole procedure to build the traditional parametric mesh is complex and the manual intervention can hardly be avoided, especially once the geometry has been changed greatly.

To create the proposed non-parametric mesh-based MRN model, selecting the proper mesh element type is the critical step at the very beginning of the modelling. A two-flux-path sector element model would be a good option for generating

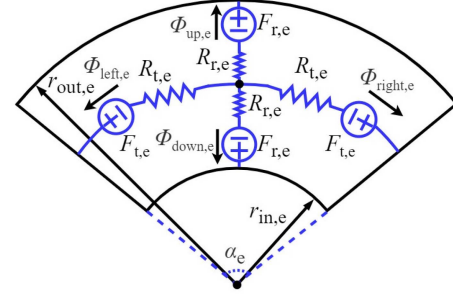


Fig. 2. A two-flux-path sector element of the non-parametric mesh-based MRN model, where $F_{r,e}$ and $F_{t,e}$ are the radial and tangential magnetomotive forces (MMFs); R_r and R_t are the radial and tangential reluctances; $\Phi_{up,e}$, $\Phi_{down,e}$, $\Phi_{left,e}$, and $\Phi_{right,e}$ are the magnetic flux components in the up, down, left, and right directions, respectively; α_e is the angular width of the sector; $r_{in,e}$ and $r_{out,e}$ are element's inner and outer radii [36].

the non-parametric mesh of radial interior PMSMs as displaced in Fig. 2. It can be seen in Fig. 2 that for each sector element, the magnetic flux travels along both the radial and tangential directions, and the corresponding radial reluctance $R_{r,e}$ and tangential reluctance $R_{t,e}$ can be defined as [36]

$$\begin{cases} R_{r,e} = \frac{1}{2\mu_0\mu_r l_{ef}} \frac{1}{\alpha_e} \ln\left(\frac{r_{out,e}}{r_{in,e}}\right) \\ R_{t,e} = \frac{1}{2\mu_0\mu_r l_{ef}} \frac{\alpha_e}{\ln\left(\frac{r_{out,e}}{r_{in,e}}\right)} \end{cases} \quad (1)$$

where μ_0 is the vacuum permeability; μ_r is the relative permeability; l_{ef} is the effective axial length.

The automatic procedure of creating the proposed non-parametric mesh for the whole physical region of a radial PMSM can be arranged as follows

- 1) Create 2-D geometry;
- 2) Select a single basic element (e.g., sector or rectangle);
- 3) Discretize the geometry by dividing the whole geometry into layers along both the radial and tangential directions so that the fundamental mesh element is a sector;
- 4) Identify the physical region of each node by the geometric outline of different areas, e.g., stator core and armature winding, thereby the physical boundary of the mesh for each region can be created automatically;
- 5) Automatic node and branch numbering;
- 6) Build mesh map and circuit topology by programs.

The non-parametric mesh example of a radial PMSM as shown in Fig. 3 with 180×80 nodes can be generated automatically, once the number of rows in the radial direction and number of columns in the tangential direction have been determined. It can be seen in Fig. 3 that there is only one single basic element (the two-flux-path sector element). So that the mesh quality is only affected by the input number of rows and columns. And the very complex geometry (e.g., stator tooth tip and permanent magnet air pocket) can also be accurately modelled by increasing the number of rows and columns if needed. Besides, no extra effects are needed to create the mesh in the air gap and it can be done normally as other parts of the machine. Please note that the automatic non-parametric meshing strategy used in

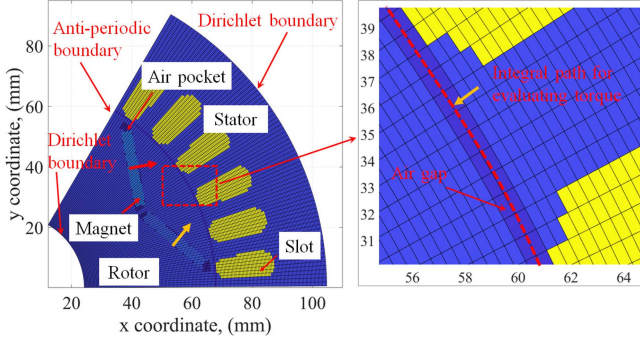


Fig. 3. A non-parametric mesh example of a 36-slot, 6-pole radial interior PMSM with 180×80 nodes for the MRN model.

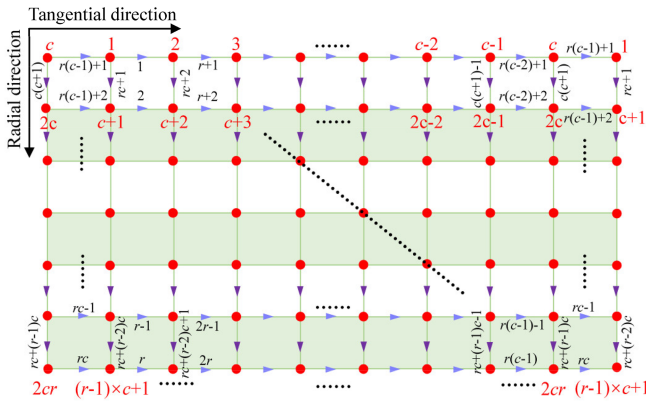


Fig. 4. Node and branch numbering of a mesh with $r \times c$ nodes, where r is the number of rows (radial direction) and c is the number of columns (tangential direction).

Fig. 3 may cause the situation that one element may contain nodes belonging to different physical regions. This case will not cause troubles for evaluating the radial reluctance $R_{r,e}$ and tangential reluctance $R_{t,e}$ as presented in Fig. 2. Because, all the reluctance components are first calculated at nodes with different permeabilities, then the branch reluctance in one element will be assembled by the nodal reluctance.

Thanks to the non-parametric mesh strategy, the automatic node and branch numbering becomes doable and finally the circuit topology can also be created accordingly. Fig. 4 shows the circuit topology of a mesh with $r \times c$ nodes, in which both the nodes and branches have been numbered as the following principle: first the tangential direction and then the radial direction. Moreover, for a mesh with $r \times c$ nodes, there are $r \times c$ tangential branches and $(r-1) \times c$ radial branches, which means that the total branch number is $(2r-1) \times c$. And the positive direction of the branch is defined as the same as the arrow direction as shown in Fig. 4.

In the end of this section, the detailed comparison of the parametric and non-parametric mesh MRNs has been listed in Table I, in which the non-parametric mesh MRN model has the advantages of good mesh quality, easy implementation, high accuracy, good repeatability, etc.

TABLE I
COMPARISON OF THE PARAMETRIC AND NON-PARAMETRIC MESH MRNs

Comparison item	Parametric mesh MRN model	Non-parametric mesh MRN model
Basic element	Multiple	Single
Mesh quality	Coarse	Dense
Air gap mesh	Complex	Simple
Topology generation	Difficult	Easy
Repeatability	Moderate	Good
Level of automation	Low	High
Accuracy	Moderate	High
Computing speed	Fast	Moderate

B. Magnetic Field Excitation Modelling

If the eddy-current effects of the magnet are neglected for a PMSM, two kinds of magnetic field excitations need to be modelled in the MRN model. One is the permanent magnet excitation, and the other is the stator winding excitation. For the permanent magnet excitation, the magnetic flux components $\Phi_{r,r}$ in the radial direction and $\Phi_{r,t}$ in the tangential direction belonging to the permanent magnet physical region can be expressed as

$$\begin{cases} \Phi_{r,r} = \frac{1}{2} B_r \cdot \cos \beta \cdot l_{ef} \cdot \frac{r_{out,e} + r_{in,e}}{2} \cdot \alpha_e \\ \Phi_{r,t} = -\frac{1}{2} B_r \cdot \cos \beta \cdot l_{ef} \cdot (r_{out,e} - r_{in,e}) \end{cases} \quad (2)$$

where B_r is the magnet remanence; β is the magnetization direction of each element.

In case of the nodes belongs to the stator winding physical region, it is not necessary to model the magnetomotive force (MMF) separately generated by a single closed coil. The three-phase synthesized MMF can be first computed and then the synthesized MMF $F_{r,MMF}$ can be assigned to the nodes in the annulus region as shown in Fig. 5(a) [37]. According to Ampere's law, the single phase normalized MMF $F(\theta_{sp})$ is numerically equal to current linkage $\Theta(\theta_{sp})$ as [38]

$$F(\theta_{sp}) = \Theta(\theta_{sp}) = \int_0^{2\pi} \frac{z_Q(\theta_{sp}) I(\theta_{sp})}{b_1} d\theta_{sp} \quad (3)$$

where b_1 is the slot opening; I and z_Q are the stator phase current and number of conductors belong to a single phase varying with spatial position θ_{sp} . The corresponding synthesized MMF can be easily obtained by calculating the sum of all phases' MMFs as shown in Fig. 5(b).

For each element in the annulus region in Fig. 5(a), the excitation MMF is arranged only in the radial direction, which is different from the modelling of the permanent magnet excitation. Besides, the annulus in Fig. 5(a) might be sliced into several rows by the non-parametric mesh. Therefore, the synthesized MMF $F_{r,MMF}$ of each element can be computed by

$$F_{r,MMF} = \frac{F_{tot}}{2r_{MMF}} \quad (4)$$

where F_{tot} is the three-phase synthesized MMF and r_{MMF} is the number of rows of the elements staying in the annulus in Fig. 5(a). Finally, the MMF excitation $F_{r,MMF}$ of each element needs to be converted into the flux excitation $\Phi_{r,MMF}$ similar to

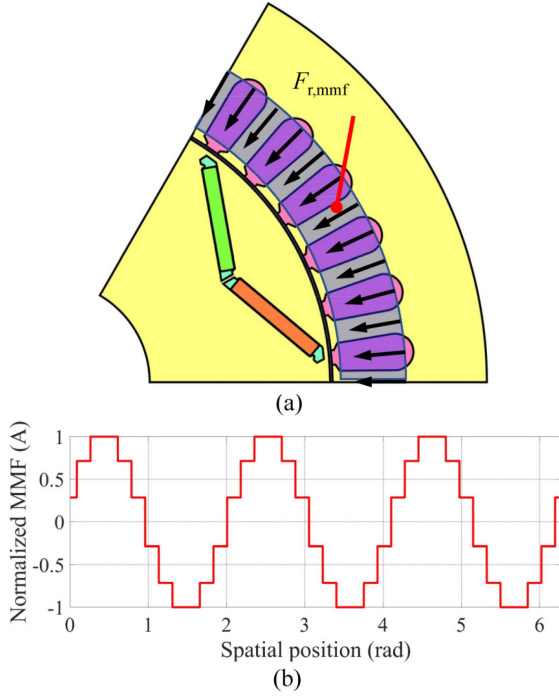


Fig. 5. The stator winding excitation modelling of a radial interior PMSM. (a) Stator winding excitation arrangement. (b) Normalized three-phase synthesized MMF waveform.

the treatment in (2), which can be further rewritten as

$$\Phi_{r,MMF} = \frac{F_{r,MMF}}{R_{r,e}} \quad (5)$$

C. System of Equations for Nodal-based Magnetic Reluctance Network Model

Nodal analysis has been used to solve the circuits based on Kirchhoff's law. The nodal permeance matrix $\mathbf{G}_n$ for a mesh with $r \times c$ nodes can be expressed as

$$\mathbf{G}_{n,j \times j} = \mathbf{A}_{j \times k} \mathbf{G}_{b,k \times k} \mathbf{A}_{j \times k}^T \quad (6)$$

where j is the number of nodes for the circuit analysis and it is equal to the number of nodes $r \times c$; k is the number of branches and it is equal to the number of branches $(2r-1) \times c$; $\mathbf{A}$ is the incidence matrix with a dimension of $j \times k$, which can be further obtained from the node and branch connection information in Fig. 4; $\mathbf{G}_b$ is a $k \times k$ diagonal matrix, which can be expressed by the branch permeance as

$$\mathbf{G}_{b,k \times k} = \text{diag} \left[\frac{1}{R_{b,1}}, \frac{1}{R_{b,2}}, \dots, \frac{1}{R_{b,k}} \right] \quad (7)$$

where $R_{b,k}$ is the k^{th} element of the branch reluctance vector $\mathbf{R}_b$, and it can be further estimated by the reluctance between the two adjacent nodes as described in (1).

The nodal magnetic scalar potential matrix $\mathbf{F}_n$ can be computed by

$$\mathbf{F}_{n,j \times 1} = \mathbf{G}_{n,j \times j}^{-1} \mathbf{A}_{j \times k} \Phi_{ns,k \times 1} \quad (8)$$

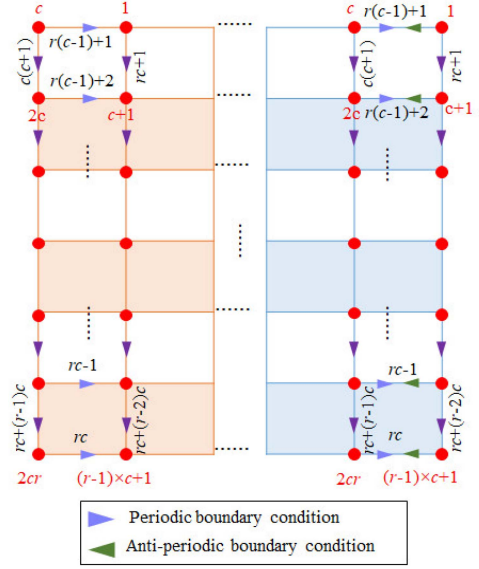


Fig. 6. Boundary conditions for the MRN model.

where Φ_{ns} is the excitation magnet flux vector for each branch as described in (2) and (5).

The magnetic voltage drop of each branch $\mathbf{F}_b$ can be further expressed as

$$\mathbf{F}_{b,k \times 1} = \mathbf{A}_{j \times k}^T \mathbf{F}_{n,j \times 1} \quad (9)$$

Then, the magnet flux of each branch Φ_b can be estimated by

$$\Phi_{b,k \times 1} = \mathbf{F}_{b,k \times 1} \oslash \mathbf{R}_{b,k \times 1} + \Phi_{ns,k \times 1} \quad (10)$$

where $\oslash$ is the Hadamard division.

D. Boundary Conditions

The Dirichlet boundary condition has already been considered automatically as shown in the circuit topology in Fig. 4. Because there are only three branches for the nodes in the upper and lower boundaries in Fig. 4, which means that the magnetic flux cannot travel out of the boundary.

Periodic and anti-periodic boundary conditions can be used to reduce the matrix dimensions, which may help to improve the computation speed significantly. The minimum unit to be modelled considering boundary conditions is $[\text{GCD}(Q_s, p)]/2$, where GCD is the greatest common divisor; Q_s is the number of stator slots; p is the number of pole pairs. Consequently, boundary conditions will be determined by the number of poles of the minimum unit, e.g., using periodic boundary for even number and anti-periodic boundary condition for odd number.

Fig. 6 shows the circuit topology for building the periodic and anti-periodic boundary conditions. It can be seen in the figure that the nodes in the last column (i.e., column c) are connected to those in the first column (i.e., column 1). For both boundary conditions, the currents always flow into nodes in the first column on left-hand side for each topology in Fig. 6. If periodic boundary condition is applied, the currents in the last column on the right-hand side need to be set as flowing

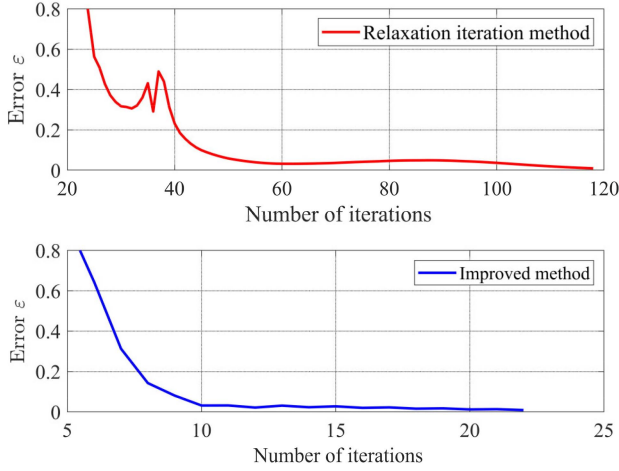


Fig. 7. Global flux density error ε as function of iterations for both the relaxation iteration method ($k_r = 0.12$) and the improved method ($d = 0.12$) for the MRN model with 200×60 nodes.

out of nodes. And the currents will flow into the nodes in the last column for anti-periodic boundary conditions. Finally, the boundary conditions can be arranged by modifying the incidence matrix $\mathbf{A}$ in (6) accordingly.

E. Nonlinear Iteration

The nonlinear B - H curve has been used for the modelling of the nonlinear permeability of the material. One potential approach is to use the relaxation iteration method, which can be arranged as follows as reported in [39] and [40]

- 1) Initialize the permeability of the nonlinear material with the vacuum permeability μ_0 ;
- 2) Solve (6)–(12) and get the nodal flux density matrix $\mathbf{B}$;
- 3) Calculate the nodal permeability matrix μ by interpolating the nonlinear B - H curve of the material with the solved nodal flux density matrix $\mathbf{B}$;
- 4) Update the nodal permeability matrix as

$$\mu^{i+1} = \mu^i + k_r (\mu - \mu^i) \quad (11)$$

where i is the number of iterations; k_r ($0 < k_r < 1$) is the relaxation factor;

- 5) Calculate the flux density matrix of the following step and estimate the global flux density error ε from all nodes as

$$\varepsilon = \|\mathbf{B}^{i+1} - \mathbf{B}^i\| / \|\mathbf{B}^i\| \quad (12)$$

- 6) Stop the iteration if maximum flux density error ε is smaller than the threshold, e.g., 0.001.

Another improved method can be obtained by substituting (11) in the i^{th} iteration with the following expression [41]

$$\mu^{i+1} = \mu^d \{\mu^i\}^{1-d} \quad (13)$$

where d ($0 < d < 1$) is the damping coefficient.

Fig. 7 shows the global flux density ε as function of iterations for relaxation iteration and improved methods. Both the relaxation factor k_r and damping coefficient d have been tuned for achieving the minimum iterations. In Fig. 7, the improved

method (22 iterations) reaches the convergence faster than the relaxation iteration method (118 iterations), therefore, the improved method has been determined to be used in this paper.

F. Rotor Movement and Solving Process

The rotor movement can be realized by interchanging the nodes in the moving region, so that no remeshing is needed for different rotor positions. Taking the nodal permeance matrix $\mathbf{G}_n$ for example, the matrix in (6) can be further expressed as

$$\mathbf{G}_n = \begin{bmatrix} \mathbf{G}_r \\ \mathbf{G}_s \end{bmatrix} \quad (14)$$

where $\mathbf{G}_s$ is the nodal permeance submatrix belonging to the stationary part; $\mathbf{G}_r$ is the nodal permeance submatrix belonging to the rotating region, and it can be written as

$$\mathbf{G}_r = \begin{bmatrix} G_{[1,c-n+1]} & \cdots & G_{[1,j]} & G_{[1,1]} & \cdots & G_{[1,c-n]} \\ G_{[2,c-n+1]} & \cdots & G_{[2,j]} & G_{[2,1]} & \cdots & G_{[2,c-n]} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ G_{[m,c-n+1]} & \cdots & G_{[m,j]} & G_{[m,1]} & \cdots & G_{[m,c-n]} \end{bmatrix} \quad (15)$$

where n is the number of the movement steps; m is the number of the rows of the matrix in the moving region. It can be seen in (15) that the movement is achieved by interchanging the nodal physical quantities. Therefore, by using this kind of movement arrangement, the simulation time step is limited to a discrete variable and the possible time step is a multiple of a minimum time step if time-stepping analysis is used.

Finally, the entire solving procedure of the MRN model can be presented in Fig. 8, in which both the nonlinear iteration and rotor movement have been considered.

III. POST-PROCESSING OF THE NON-PARAMETRIC MESH MAGNETIC RELUCTANCE NETWORK MODEL

A. Magnetic Flux Density

According to (6)–(10) and Fig. 1, the flux density vectors $\mathbf{B}_{r,n}$ in the radial direction and $\mathbf{B}_{t,n}$ in the tangential direction of each node can be calculated as

$$\begin{cases} B_{r,n} = \sqrt{\frac{\Phi_{up,e}^2 + \Phi_{down,e}^2}{2(l_{ef} \cdot \frac{r_{out,e} + r_{in,e}}{2} \cdot \alpha_e)^2}} \cdot \text{sgn}\left(\frac{\Phi_{up,e} + \Phi_{down,e}}{2}\right) \\ B_{t,n} = \sqrt{\frac{\Phi_{left,e}^2 + \Phi_{right,e}^2}{2[l_{ef} \cdot (r_{out,e} - r_{in,e})]^2}} \cdot \text{sgn}\left(\frac{\Phi_{left,e} + \Phi_{right,e}}{2}\right) \end{cases} \quad (16)$$

where sgn is the sign function.

Consequently, the corresponding flux density magnitude B_n of each node can be finally obtained from

$$B_n = \sqrt{B_{r,n}^2 + B_{t,n}^2} \quad (17)$$

B. Magnetic Flux Linkage and Back EMF

The magnetic flux linkage simulated by the MRN model can be calculated by the integral of the radial flux density $B_{r,n}$ in the

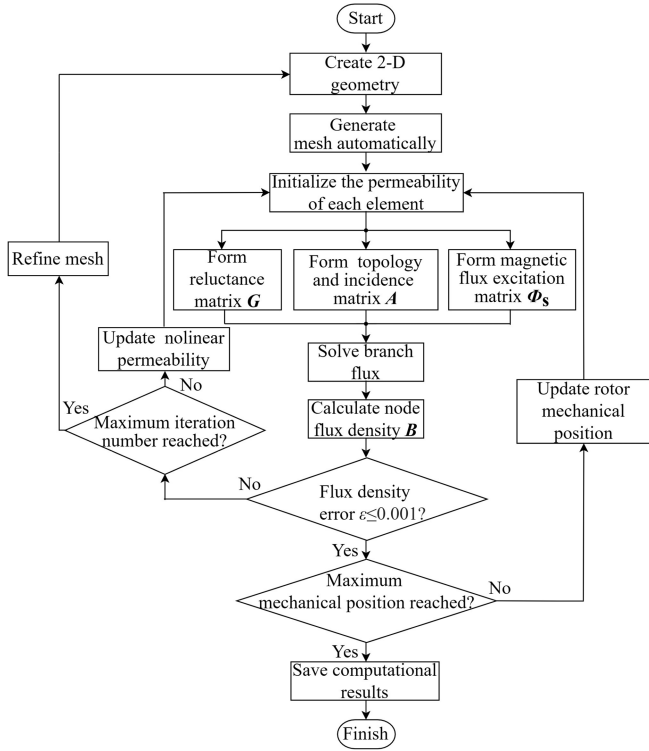


Fig. 8. Flow chart of solving the MRN model considering both the nonlinear iteration and rotor movement.

middle of the air gap as

$$\psi = N_c r_{ave} l_{ef} \int_{\theta_0}^{\theta_0 + \theta_c} B_{r,n} d\theta_{sp} \quad (18)$$

where N_c is the number of turns in series per phase; r_{ave} is the average air-gap radius; θ_0 is the stator winding initial spatial angle; θ_c is the coil pitch measured by the mechanical degree.

The back EMF varying with time t can be further estimated by the differential of the flux linkage as

$$E(t) = -\frac{d\psi(t)}{dt} \quad (19)$$

According to (18) and (19), once the radial flux density component has been solved, the flux linkage and back EMF can be easily deduced. It has to be emphasized here that the logic to solve flux linkage by FEA is different from (18) used by MRN and it can be expressed as [42]

$$\psi = \frac{N_c l_{ef}}{S} \left(\iint_{S_+} A dS - \iint_{S_-} A dS \right) \quad (20)$$

where A is the magnetic vector potential; S_+ and S_- are the positively oriented and negatively oriented winding regions modelled in FEA.

C. Electromagnetic Torque

There are two main methods for solving the electromagnetic torque for the MRN and FEA. One is the Maxwell stress tensor method, which has been widely used for the analytical modelling

of electrical machines and it can be expressed as [43]

$$T_e = \frac{l_{ef}}{\mu_0} r_{ave}^2 \int_0^{2\pi} B_{r,n}(t) B_{t,n}(t) d\theta_{sp} \quad (21)$$

The integral for evaluating the torque described in (21) has been conducted in the middle of the air gap between the stator and rotor, and the specific path has been highlighted in Fig. 3. The other method is the virtual work method, which is preferred by FEA and it can be written as

$$T_e = -\frac{\partial W_m}{\partial \theta_{sp}} \quad (22)$$

where W_m is the magnetic energy stored in the air-gap region, which can be further expressed as

$$\begin{cases} W_m = \int_V \frac{1}{2} \mu_0 H^2 dV \\ H = \frac{B}{\mu_0} = \frac{1}{\mu_0} \nabla \times A \end{cases} \quad (23)$$

where V is the region refers to the air gap between the stator and rotor in FEA, in which the volume integral has been applied; H is the magnetic strength in the air-gap region.

D. Iron Losses

Owing to the non-parametric mesh MRN model, the iron loss density $P_{Fe,e}$ for each element can also be derived from the flux density magnitude B_n in (17) as

$$P_{Fe,e} = \sum_{\nu=1}^N \left[k_h f_{\nu} B_{n,\nu}^2 + k_c (f_{\nu} B_{n,\nu})^2 + k_e (f_{\nu} B_{n,\nu})^{1.5} \right] \quad (24)$$

where k_h is the hysteresis loss coefficient; k_c is the classical loss coefficient; k_e is the excess loss coefficient; ν is the order of the high-frequency harmonic.

In (24), iron losses are computed over one electrical time period, and flux density harmonics will be obtained by FFT in mesh nodes in the lamination region. Consequently, the total iron losses can be calculated by the integral of the iron loss density $P_{Fe,e}$ in the lamination region.

In FEA the iron loss density $P_{Fe,e}$ for each element is calculated as [44]

$$\begin{aligned} P_{Fe,e} = & k_h f B_{n,max}^2 + k_c f \int_0^T \left(\frac{dB_n}{dt} \right)^2 dt \\ & + k_e f \int_0^T \left| \frac{dB_n}{dt} \right|^{1.5} dt \end{aligned} \quad (25)$$

where T is the time for the integral.

In (24) and (25), the loss coefficients k_h , k_c , and k_e are 497.31 W/m³, 0.24 W/m³, and 1.96 W/m³, respectively, estimated by fitting the multi-frequency B - P curves of the lamination (50W600) as shown in Fig. 9. The coefficients are optimum results, when the minimum square of the difference between the fitted curves and the tested curves is achieved.

E. Copper and Mechanical Losses

To predict the permanent magnet losses, the induced eddy currents need to be modelled by the MRN, which has not been

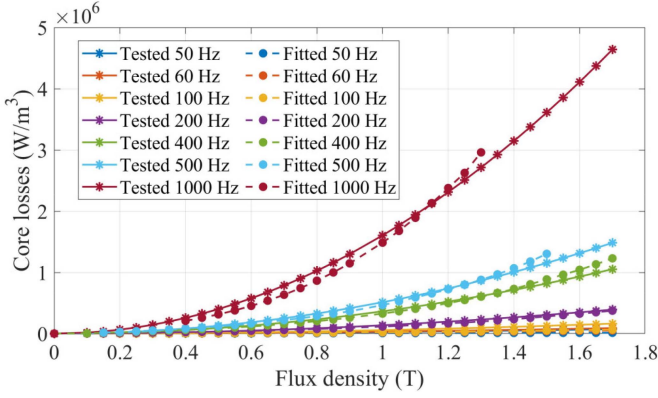


Fig. 9. Multi-frequency B - P curve fitting for the estimation of iron loss coefficients.

TABLE II
MAIN PARAMETERS OF THE PMSM

PARAMETER	Value
Rated power (kW)	25
Rated voltage (V)	144
Rated current (A)	152
Rated torque (Nm)	150
Rated speed (rpm)	1600
Number of poles	6
Number of stator slots	36
Stator outer diameter (mm)	210
Stator inner diameter (mm)	136
Rotor outer diameter (mm)	134.8
Rotor inner diameter (mm)	48
Stack length (mm)	105
Magnet grade	N38SH
Magnet remanence (T)	1.18
Lamination grade	50W600

accomplished yet and this aspect is not the main concern of this paper. The copper losses P_{Cu} for the FEA and MRN models are both estimated by the analytical approach as

$$P_{Cu} = 3I^2 \frac{N_c l_{av}}{\sigma S_c} \quad (26)$$

where l_{av} is the average length per turn; σ is the conductivity; S_c is the cross-sectional area of the wire.

The mechanical losses P_ρ are predicted by the empirical formula as [45]

$$P_\rho = k_\rho D_r (l_r + 0.6\tau) v_r^2 \quad (27)$$

where k_ρ is the empirical coefficient; D_r and l_r are the rotor outer diameter and length; τ is the pole pitch; v_r is the rotor peripheral speed.

IV. VERIFICATION BY FINITE ELEMENT ANALYSIS AND EXPERIMENTS

In this section, to verify the proposed non-parametric mesh MRN model, detailed comparisons of an interior 36-slot, 6-pole PMSM shown in Fig. 3 have been made to the FEA and experimental results, and the main parameters of the PMSM are listed in Table II. For sake of simplicity, 1/6 model with

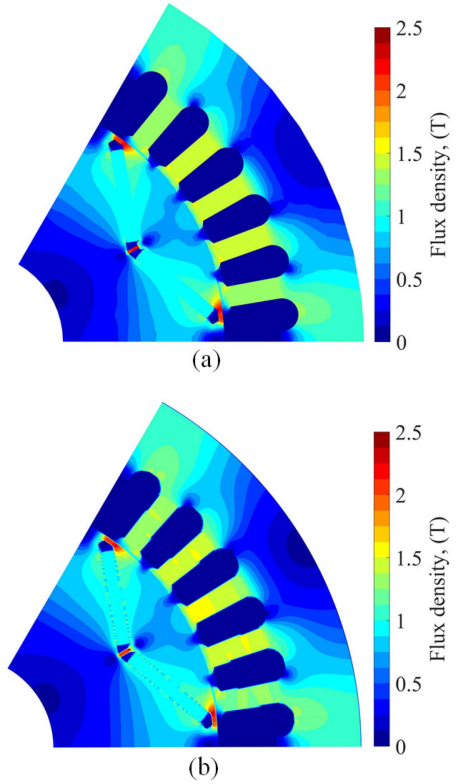


Fig. 10. No-load magnetic flux density distribution. (a) Solved by FEA. (b) Solved by MRN.

anti-periodic boundary conditions has been modelled in both FEA and MRN. And Dirichlet boundary conditions have been applied in the stator inner and rotor outer boundaries for both methods. Besides, it has to be emphasized here that a two-layer distributed winding with a short pitch ($y = 5/6$) has been used in the modelling.

A. No Load Verification

The global physical quantities such as magnetic flux density distribution might be a good option for the comparison of the computational accuracy between the FEA and MRN. Fig. 10 shows the no-load magnetic flux density distribution solved by both methods. It can be seen in the figure that the magnetic field distribution solved by the MRN are highly similar to that solved by FEA. Even the local magnetic flux density around the air pocket is about the same, which means that the magnetic field quantities solved by non-parametric mesh MRN model should be reliable.

As presented in (21), the air-gap flux density is crucial for calculating the electromagnetic torque by the Maxwell stress tensor method. Fig. 11 shows both the radial and tangential no-load air-gap flux density components as function of mechanical position solved by both the FEA and MRN. It can be seen in the figure that the radial flux density solved by MRN accords well with that solved by FEA, which also indicates that the flux linkage solved by (18) will be correct.

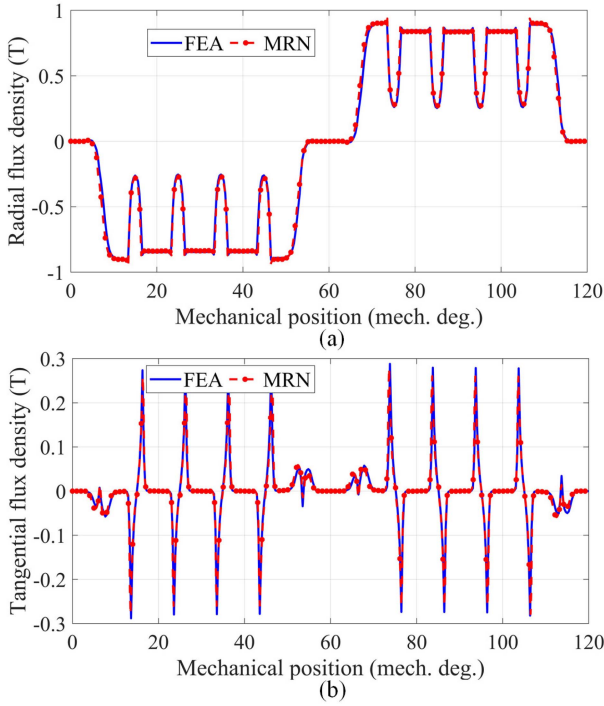


Fig. 11. No-load air-gap flux density as function of mechanical position solved by both the FEA and MRN. (a) Radial air-gap flux density. (b) Tangential air-gap flux density.

In Fig. 12, the almost same flux linkage waveforms have been obtained by FEA and MRN using different logics for modelling and post-processing, which means that the proposed MRN might be a good substitution for FEA in some applications. It has to be emphasized here that (18) is used to estimate the flux linkage and the integral in (18) needs to be conducted for each coil because of the usage of the two-layer distributed short pitch winding, which means that 4 individual integrals with different paths will be completed for each phase.

Besides, the back EMFs without considering the skewing effects deduced from the differential of the flux linkage by these two methods also show the similar waveform as displaced in Fig. 12(b). Owing to the non-parametric mesh property, the ripple component caused by slot opening or reluctance variation can be accurately captured by the proposed MRN model, which might be the main advantage of the proposed MRN compared to other analytical approaches. Finally, by using the traditional multi-slice method, both simulated back EMF waveforms considering stator skewing effects have been compared to the measured results. And the fundamentals are about 53.33 V, 52.44 V, and 53.86 V for the FEA, MRN and measurement, respectively.

The no-load cogging torque is another import index for checking the validity of the proposed method and the corresponding waveforms are compared in Fig. 13. One electrical time period (120 mech. deg.) is simulated for both cases in Fig. 13, which means that the cogging torque order should be 12. Because, theoretically the cogging torque order will be $\text{LCM}(36, 6) = 36$ for 360 mech. deg., where LCM means the least common multiple. Moreover, the peak-to-peak values of the torque in

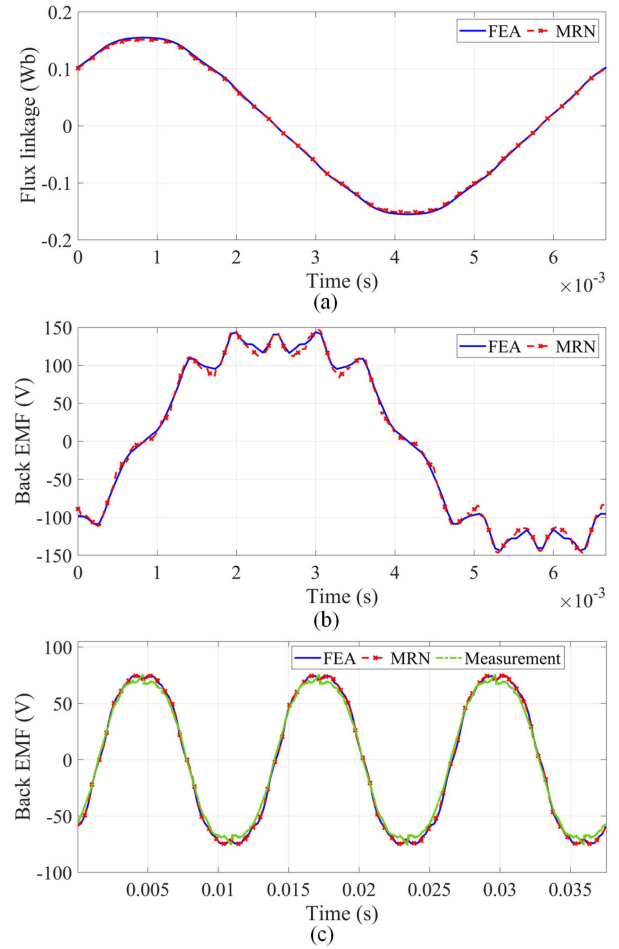


Fig. 12. No-load flux linkage and back EMF waveforms at 1600 rpm estimated by the FEA, MRN models and measurement. (a) Flux linkage. (b) Back EMF without considering skewing effects. (c) Back EMF considering skewing effects (the stator slot is skewed by one stator slot pitch and it has been considered by the traditional multi-slice method in both the FEA and MRN).

Fig. 13 are 10.21 Nm by FEA and 10.34 Nm by MRN and the error is about 1.27%.

In this section, the no load comparison has been made in detail. And it has been noticed that the proposed MRN model can provide accurate results as the FEA does including both the field quantities and deduced electromagnetic performances.

B. Load Verification

The maximum torque per ampere (MTPA) control strategy has been applied for the load simulation because the motor studied in this section is an interior PMSM. Fig. 14 shows the magnetic flux distributions at 20 A by the MTPA control strategy solved by both the FEA and MRN. It can be seen in the figure that the both distributions show almost the same details except some small difference on the stator tooth area in the map plotted by MRN. And the error comes from the truth that the equivalent current i.e., MMF has been applied in the stator region as described in Fig. 5 by MRN.

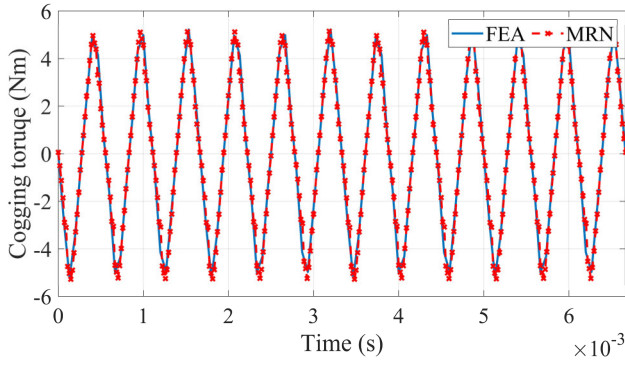


Fig. 13. No-load cogging torque waveforms at 3000 rpm solved by both the FEA (virtual work method) and MRN (Maxwell stress tensor method).

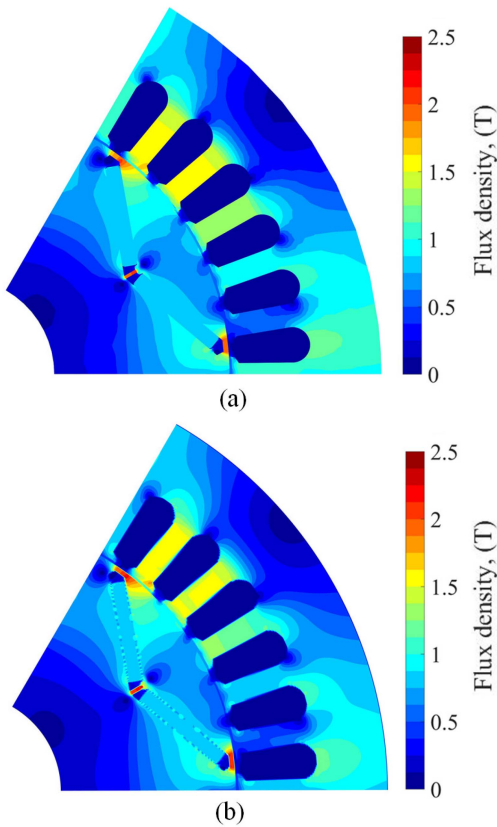


Fig. 14. Magnetic flux density distribution at 20 A by the MTPA control strategy. (a) Solved by FEA. (b) Solved by MRN.

Fig. 15 shows the details about the air-gap flux density spatial distributions at 97 A by the MTPA control strategy. It can be noticed that the tangential components solved by these two methods accord well with each other. However, some distortions can be observed in the radial component waveforms, especially in the ranges from 26 to 33 mech. deg. and 86 to 93 mech. deg. These errors might result from the equivalent MMF setting from the current excitation as shown in Fig. 5(b). It can be seen in Fig. 5(b) that the MMF for the MRN is assumed to be concentrated on some part of the geometry, to be more specific, the stator teeth, which means that the MMF in MRN is an

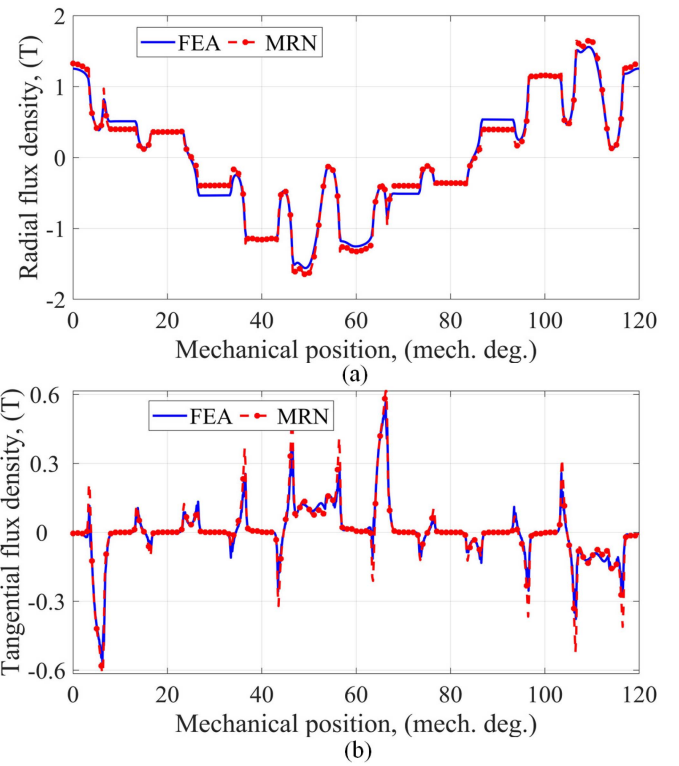


Fig. 15. Air-gap flux density as function of mechanical position solved by both the FEA and MRN at 97 A by the MTPA control strategy. (a) Radial air-gap flux density. (b) Tangential air-gap flux density.

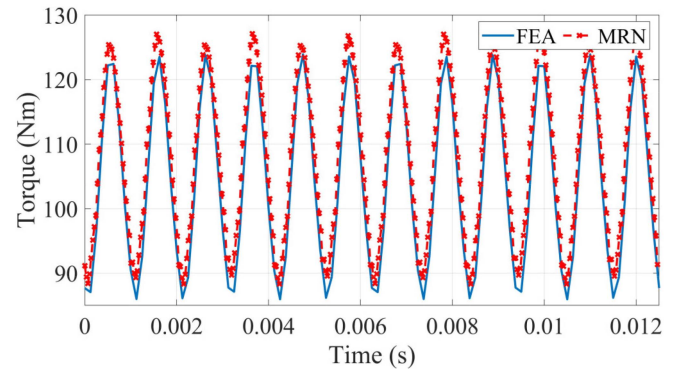


Fig. 16. Torque waveforms at 97 A, 1600 rpm by the MTPA control strategy solved by both the FEA (virtual work method) and MRN (Maxwell stress tensor method).

equivalent lumped quantity. However, theoretically, the MMF should be distributed in the whole region based on the right-hand rule, which means that in reality the MMF should be a distributed quantity.

Despite some small errors in the air-gap flux density distribution, the MRN still provides precise torque waveforms in load condition, which have been compared to those solved by FEA in Fig. 16. The average torque is about 104.30 Nm and 107.85 Nm by MRN and FEA, respectively, which results in an error of 3.40%. The peak-to-peak torque values are about 37.97 Nm and 38.78 Nm and the corresponding error is about 2.12%.

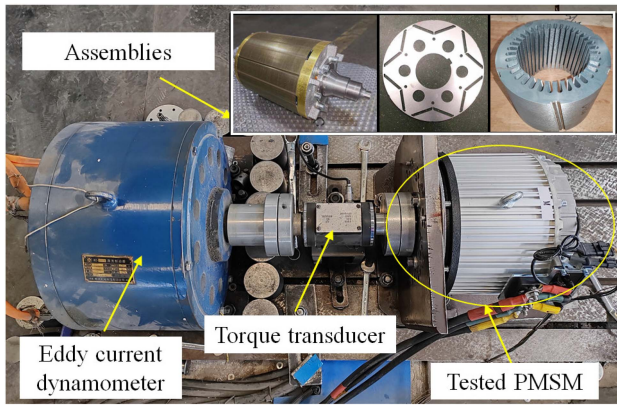


Fig. 17. The load test bench for the on-load measurement.

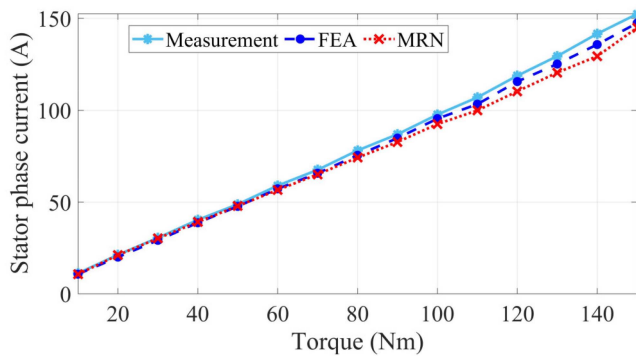


Fig. 18. Comparison of the tested and simulated stator phase currents in constant speed 1600 rpm different load conditions by the MTPA control strategy.

Therefore, it can be noticed that the proposed non-parametric mesh MRN model can not only provide reliable average torque, but also precise torque waveforms in load condition.

Fig. 17 shows the test bench for the on-load measurement and the motor assemblies have also been displaced in the figure. Fig. 18 shows the comparison of the stator phase currents varying with the torque at 1600 rpm estimated by both experiments and simulations. It can be seen in this figure that the simulated results accord quite well with the experimental results in the low load conditions from 10 Nm to 80 Nm. The error becomes higher with the increment of the torque and the maximum errors at 140 Nm are 4.12% by FEA and 8.62% by MRN with respect to the tested results. From the simulation point view, the maximum error is also achieved at 140 Nm and it is 4.68% by MRN with respect to the FEA simulation results.

The terminal voltage might also be a good index for evaluating the accuracy of the proposed method. Fig. 19 shows the tested and simulated stator phase voltages in different conditions. The maximum errors are 4.20% by FEA at 150 Nm and 4.83% by MRN at 30 Nm with respect to the tested results. These errors should be acceptable for most industrial applications.

Owing to the non-parametric mesh property, the magnetic flux density can be obtained by the proposed MRN so that the iron loss can also be estimated by (24). Fig. 20 shows the iron losses in different load and speed conditions simulated by both the FEA

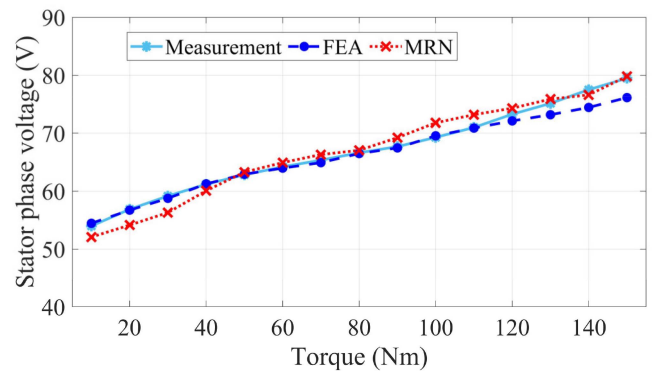


Fig. 19. Comparison of the tested and simulated stator phase voltages in constant speed 1600 rpm different load conditions by the MTPA control strategy.

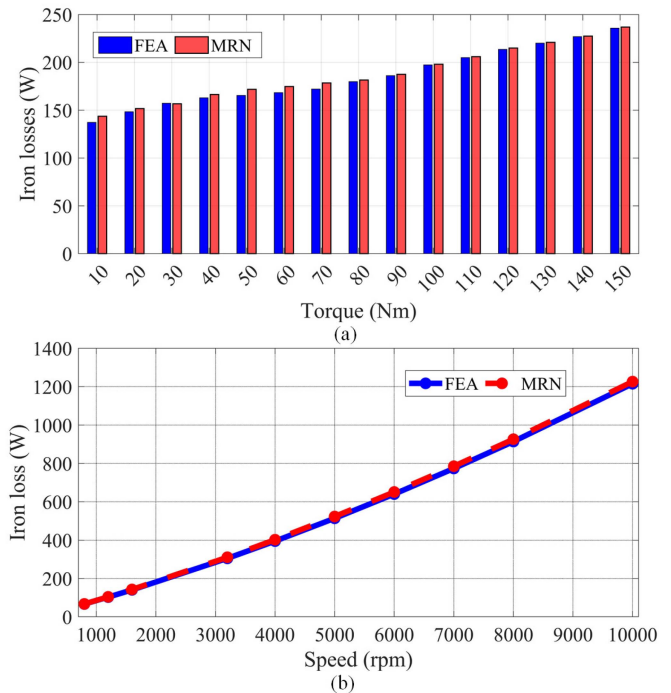


Fig. 20. Comparison of the simulated iron losses in different load and speed conditions. (a) Iron losses in different load conditions at 1600 rpm. (b) Iron losses under different operational speeds at 10 Nm.

and MRN. It can be seen in Fig. 20 that both simulations show similar results despite some small errors, and the differences may be resulted from the different mesh strategies. Consequently, the machine's efficiency can be obtained by both the experiment and simulation as shown in Fig. 21, in which the main error might come from the additional losses, permanent magnet losses and also the temperature rise.

The parallel computing speed of the proposed MRN has also been studied in detail and the corresponding simulation time has been listed in Fig. 22(damping coefficient for nonlinear iterations has been selected as 0.02 and it has not been fully optimized). It can be seen in this figure that the simulation time (40000 nodes and 100 time steps) can be greatly reduced to 230 s and 118 s by 4-core and 14-core parallel computing. Besides,

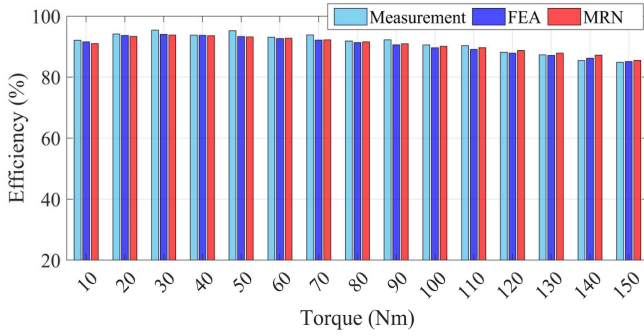


Fig. 21. Comparison of the tested and simulated efficiency in constant speed 1600 rpm different load conditions by the MTPA control strategy.

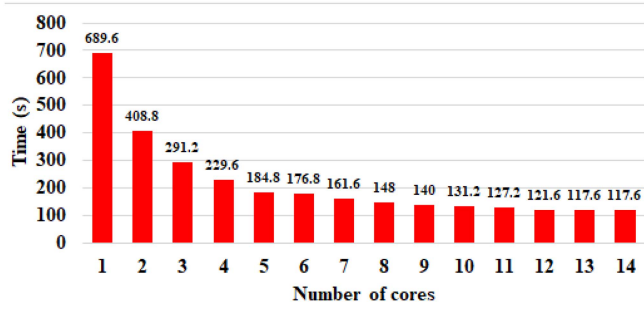


Fig. 22. Comparison of parallel computing speed of the proposed MRN (40000 nodes) and FEA (40012 nodes) with 100 time steps solved by Intel i9-10940X, 64GB. Damping coefficient for nonlinear iterations is selected as 0.02 and it has not been fully optimized.

it can also be noticed that if the damping coefficient is tuned in the proper way, the parallel computing speed can be further improved, which will be discussed in detail in the next section.

V. DISCUSSION

A. Damping Coefficient Search Algorithm

In Section II, the relaxation iteration method and the improved method (both of them can be classified into the fixed-point iteration method), have been used for the nodal-based MRN. In Fig. 7, a damping coefficient 0.12 has been found by the trial-and-error method to reach the fastest convergence with about 22 iterations. However, the trial-and-error method sometimes needs manual intervention, which means that the convergence cannot be always guaranteed if the model's geometry, mesh, and load condition are modified.

To solve this problem, a search algorithm has been applied to search a proper damping coefficient for the MRN proposed in this paper, which is originally used to find the proper relaxation coefficient for better convergence in the Newton-Raphson iteration in FEA as discussed in [46] and [47]. The damping coefficient d for each iteration can be searched by the following principle

$$d = \frac{1}{2^n}, n = 1, 2, \dots, 10 \quad (28)$$

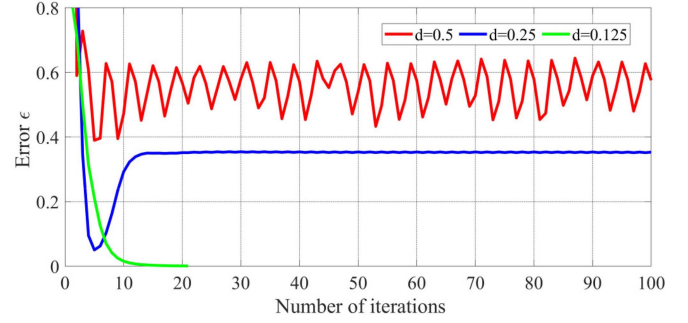


Fig. 23. Global flux density error ϵ as function of iterations by using different damping coefficients for a mesh with 200×60 nodes.

The search algorithm will be continued until reaching the maximum number of iterations or the following condition has been satisfied in each iteration

$$\epsilon^{i+1} \leq \epsilon^i \quad (29)$$

The condition described in (29) enforces the error to decrease in each iteration, which finally ensures the convergence of the system of equations. Fig. 23 can be obtained by following the search algorithm described in (28) and (29). It can be seen in the figure that the damping coefficient d drops from 1 to 0.125, and the search algorithm will stop automatically when d reaches 0.125 with 22 iterations.

To further check the validity of the search algorithm, simulation cases with different number of mesh nodes and stator phase currents have been solved by the proposed MRN with the damping coefficient search algorithm. It can be seen in Fig. 24 that the convergence has been guaranteed in all cases and the damping coefficient varies between 0.125 and 0.0625 while the number of iterations ranges from 14 to 31.

Although the damping coefficient search algorithm can ensure the convergence during the nonlinear iteration, the total simulation time may be increased by a lot if it has been used in every time step in the time-stepping analysis. Therefore, special efforts have also been paid to accelerate the convergence problem in the time-stepping analysis. The converging speed in the next time step can be further accelerated by replacing the initial condition with the physical quantities solved by the previous time step [48]. Fig. 25 shows the number of iterations in time-stepping analysis for a mesh with 40000 nodes. The damping coefficient is 0.125 found by the search algorithm for the nonlinear iteration, and the stator phase current is 60 A. It can be seen in the figure that the convergence can be achieved within 5 iterations after the second step. The simulation time is reduced significantly from 689.6 s shown in Fig. 22 to 58.4 s for the single-core case without taking the damping coefficient search time into consideration, and the total simulation time is about 80 s if the damping search time is also accounted.

Please note that the damping search algorithm is used only in the first time step in the time stepping analysis in this paper. The damping coefficient is kept as a constant after the first time step, because both the movement and excitation are not changing too much in the following time step. And also, the

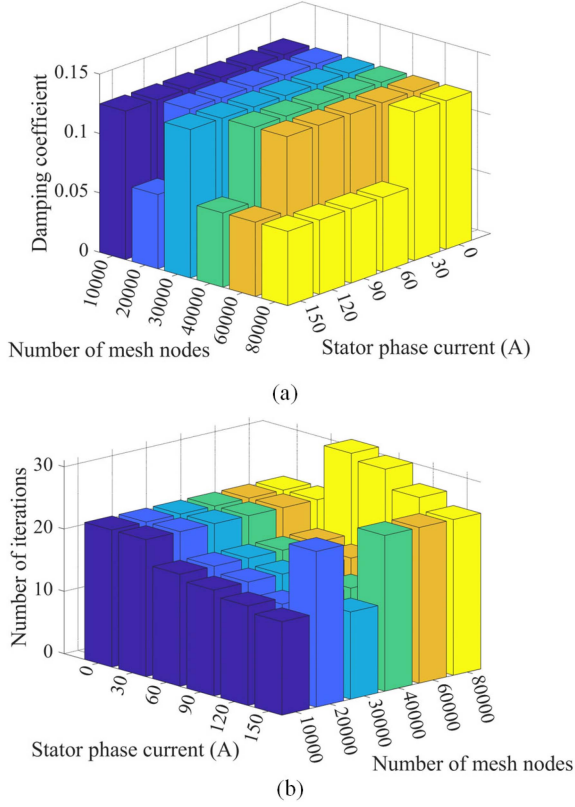


Fig. 24. Convergence details for cases with different number of mesh nodes and stator phase currents by using the damping coefficient search algorithm. (a) Damping coefficient varies with mesh nodes and stator phase currents. (b) Number of iterations varies with mesh nodes and stator phase currents.

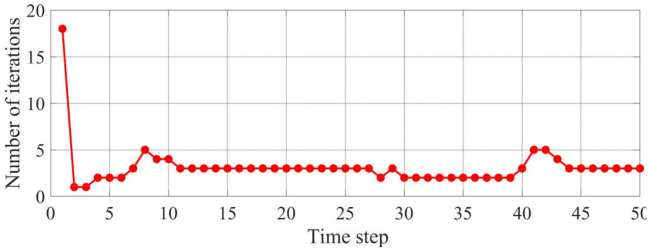


Fig. 25. Number of iterations in time-stepping analysis for a mesh with 40 000 nodes (damping coefficient 0.125, stator phase current 60 A). Please note that the damping search algorithm is used only in the first time step in the time stepping analysis, and the damping coefficient is kept as a constant after the first time step.

initial conditions are inherited from the previous time step, which would further accelerate the convergence. Consequently, the number of iterations to reach the convergence is less than five from the second step without searching the damping coefficient again as shown in Fig. 25.

The comparison of the parallel computing speed of the proposed MRN and FEA as shown in Fig. 26 can also be updated with the algorithm that the initial condition is taken from the previous time step. In Fig. 26 the simulation time approaches the minimum around 35 s by using the 5-core simulation, which is much faster compared to the simulation time 177 s

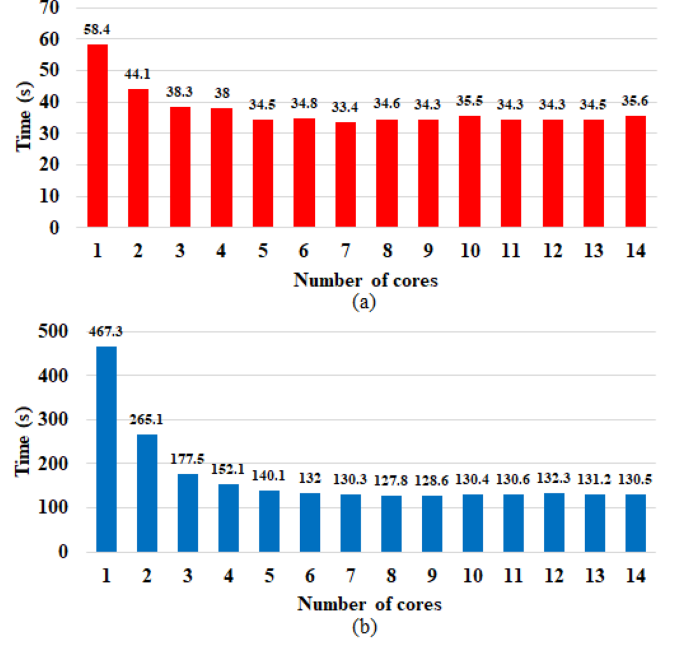


Fig. 26. Comparison of parallel computing speed of the proposed MRN (40000 nodes) and FEA (40012 nodes) with 100 time steps solved by Intel i9-10940X, 64 GB. (a) Solved by the proposed MRN (damping coefficient search algorithm has been used and the initial condition solved by the previous time step has been used). (b) Solved by FEA.

in Fig. 22 without tuning the damping coefficient and proper initial condition. Furthermore, by comparing Fig. 26(a) and (b), it can be noticed that the simulation time by both the MRN and FEA reaches the minimum if more than 4 cores are used. And the MRN is faster compared to the FEA for almost the same number of mesh nodes.

B. Newton-Raphson Iteration

Actually, there are two main methods to solve the MRN model based on the Kirchhoff's law. One is the nodal-based approach by using KCL which has been mainly used in this paper, and the other is the loop-based approach by using KVL. To improve the convergence performance during the nonlinear iteration, theoretically, the Newton-Raphson method might be applied for both cases.

The nodal analysis equations can be further transferred into the residual function vector $\mathbf{f}_n$ as

$$\mathbf{f}_{n,j \times 1} = \mathbf{G}_{n,j \times j} \mathbf{F}_{j \times 1} - \Phi_{n,j \times 1} \quad (30)$$

where Φ_n is the excitation magnet flux vector for each node; $\mathbf{F}$ is the unknown matrix. Consequently, the Newton-Raphson format for the i^{th} iteration for nodal analysis can be obtained by transferring (30) into the following equation

$$\mathbf{F}_{j \times 1}^{i+1} = \mathbf{F}_{j \times 1}^i - \frac{\mathbf{f}_{n,j \times 1}^i}{\mathbf{J}_{n,j \times j}^i} \quad (31)$$

where $\mathbf{J}_n$ is the Jacobian matrix. In [20] an analytical Jacobian matrix has been derived for the nodal-based MRN, which can

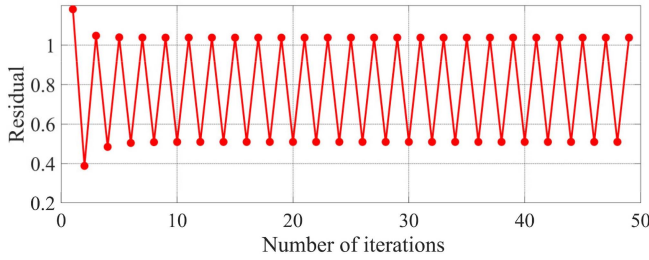


Fig. 27. Residual (error ε) during the Newton-Raphson iteration for nodal-based MRN.

be expressed as

$$\mathbf{J}_{n,j \times j} = \mathbf{A}_{j \times k} \mathbf{G}'_{b,k \times k} \mathbf{A}_{j \times k}^T \quad (32)$$

where $\mathbf{G}'_b$ can be obtained by replacing the branch reluctance in (7) with the differential branch reluctance.

Fig. 27 shows the residual during the Newton-Raphson iteration for the nodal-based MRN. The convergence has not been achieved in Fig. 27, which shows the same divergence result as reported in [49] and [33]. The reason for the divergence is quite complex. It has been emphasized in [49] that the Newton-Raphson method is not very suitable for nodal analysis, because the Jacobian matrix is often ill-conditioned. It has also been found in [33] that once the iteration goes into the nonlinear range, the Newton-Raphson iteration can hardly converge as well because of the ill-conditioned Jacobian matrix. In addition, [32] insists that the convergence of the Newton-Raphson iteration in nodal analysis is much more sensitive to the initial condition than the loop analysis and it is a very difficult task to find the proper initial condition for different meshes and load conditions. Therefore, the nodal analysis is less competitive than the loop analysis if Newton-Raphson method has to be used.

The situation becomes different when it comes to the loop-based MRN. The loop analysis equations can be transferred into the residual function vector $\mathbf{f}_l$ as

$$\mathbf{f}_{l,l \times 1} = \mathbf{R}_{l,l \times l} \Phi_{l \times 1} - \mathbf{F}_{l,l \times 1} \quad (33)$$

where l is the number of circuit loops; $\mathbf{F}_l$ are the excitation MMF for each loop; $\mathbf{R}_l$ is the loop reluctance matrix, which can be further expressed as

$$\mathbf{R}_{l,l \times l} = \mathbf{C}_{l \times k} \mathbf{R}_{b,k \times k} \mathbf{C}_{l \times k}^T \quad (34)$$

where $\mathbf{C}$ is the loop matrix.

The Newton-Raphson format for loop analysis can be further written as

$$\Phi_{l \times 1}^{i+1} = \Phi_{l \times 1}^i - \frac{\mathbf{f}_{l,l \times 1}^i}{\mathbf{J}_{l,l \times l}^i} \quad (35)$$

where $\mathbf{J}_l$ is the Jacobian matrix for the loop-based approach.

The derivation and assembly of the loop Jacobian matrix is relatively complex as presented in [15] and [33]. A simpler form of Jacobian matrix has been derived from the differential reluctance as discussed by [32]. The details about the how to transfer the Jacobian matrix provided by [15] into the differential reluctance form is shown in the Appendix, and the Jacobian

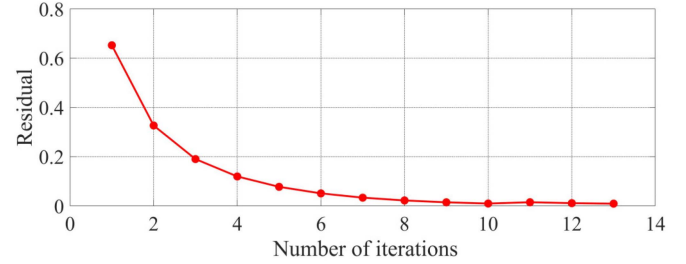


Fig. 28. Residual (error ε) during the Newton-Raphson iteration for the loop-based MRN (initial relative permeability is 1000).

TABLE III
CPU TIME PER ITERATION IN NODAL AND LOOP ANALYSIS

Nodes	Nodal analysis CPU time (s)	Loop analysis CPU time (s)
5000	0.019	0.216
10 000	0.034	0.432
15 000	0.047	0.641
20 000	0.062	0.864
25 000	0.074	1.102
30 000	0.091	2.767
35 000	0.106	3.192
40 000	0.138	3.654
60 000	0.193	5.931
80 000	0.285	7.943

matrix can be expressed as

$$\mathbf{J}_{l,l \times l} = \mathbf{C}_{l \times k} \mathbf{R}'_{b,k \times k} \mathbf{C}_{l \times k}^T \quad (36)$$

where $\mathbf{R}'_b$ is the differential branch reluctance.

Fig. 28 shows the residual during the Newton-Raphson iteration for the loop-based MRN. And the convergence has been achieved at the 13th iteration (the initial relative permeability is set as 1000), which needs less iterations compared to the nodal analysis by the fixed-point method with the proper damping coefficient shown in Fig. 25. Besides, if the initial relative permeability is set close to the knee point at 1.4 T on B - H curve as 200, the iteration takes only 6 steps.

However, the truth is that the loop-based MRN with the Newton-Raphson iteration method needs more CPU time compared to the nodal-based MRN with the fixed-point iteration method. Fig. 29 visualizes the sparsity patterns of matrix for a mesh with 200×200 nodes for both approaches. It can be seen in Fig. 29 that the loop matrix $\mathbf{C}$ in loop analysis contains much more non-zero elements than those in the incidence matrix $\mathbf{A}$ in nodal analysis. And consequently, the CPU time per iteration for the loop-based MRN is 3.654 s, while it is 0.138 s for the nodal-based MRN. Please note that no for loops are used in the calculation of the incidence matrix $\mathbf{A}$ and loop matrix $\mathbf{C}$, and sparse matrix storage has already been adopted to improve the calculation speed. And both the matrices are estimated by sparse matrix multiplications directly.

Table III lists the detailed CPU time per iteration for both methods, and it reveals that the time difference between two methods becomes even bigger with the increment of the number of mesh nodes. In addition, the CPU time per iteration in loop analysis is relatively long for the case with a large number

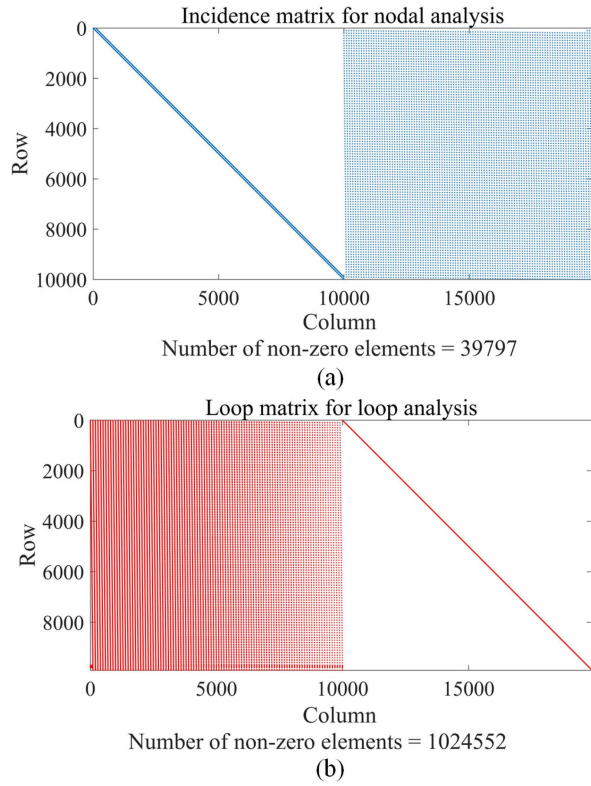


Fig. 29. Visualized sparsity patterns of matrix for a mesh with 200×200 nodes. (a) Incidence matrix A for the nodal-based MRN. (b) Loop matrix C for the loop-based MRN.

of nodes, which might not be acceptable during the practical usage. In addition, the simulation speed of the nodal-based MRN with the fixed-point iteration is relatively fast after the damping coefficient search algorithm, proper initial condition, and parallel computing have been applied. This is the main reason that the nodal-based MRN with the fixed-point iteration has been finally used in this paper.

C. Mesh Size Effects on the Accuracy and Simulation Speed

Simulations with different combinations of mesh settings have been done to further investigate the impact of the mesh size on the accuracy and simulation speed. For sake of simplicity, the effects of the numbers of radial and tangential nodes are discussed separately.

Fig. 30 shows the simulated torque and computing speed as function of number of radial nodes by the proposed nodal-based MRN with 100 time steps. Please note that number of tangential nodes is kept as a constant of 100, and the reference average torque and peak-to-peak torque are 100 Nm and 37.53 Nm solved by FEA. In Fig. 30, the computing time by single-core simulation almost increases linearly with the increment of the number of mesh nodes, and the computing time reaches about 205 s for a mesh with 800×100 nodes. Besides, it can also be seen in this figure that the accuracy of the average and peak-to-peak torque can be guaranteed if more than 300 radial nodes are used. And the corresponding simulation time for the case (300×100

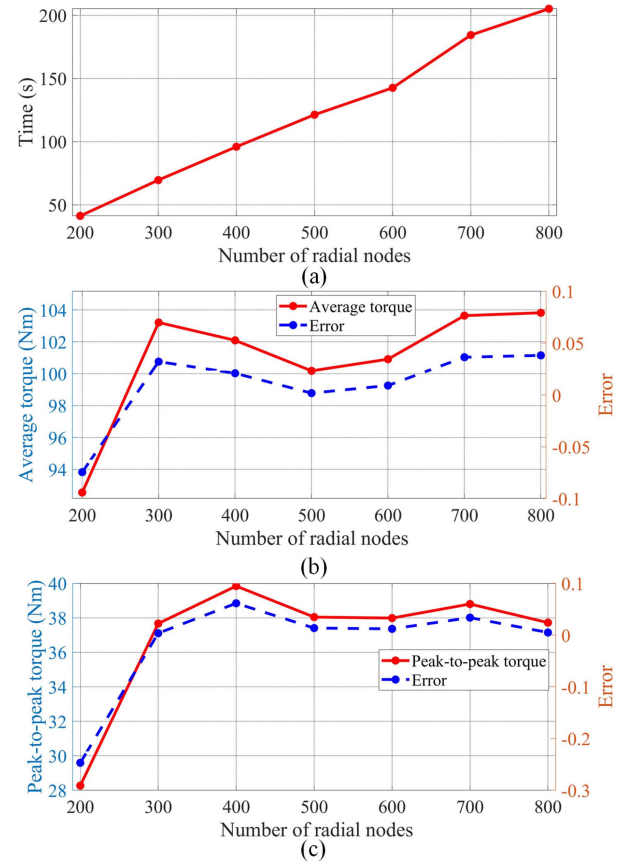


Fig. 30. Simulated torque and computing speed as function of number of radial nodes by the proposed nodal-based MRN with 100 time steps. (the number of tangential nodes is kept as a constant of 100, and the reference average torque and peak-to-peak torque are 100 Nm and 37.53 Nm solved by FEA). (a) Computing time by single core. (b) Average torque. (c) Peak-to-peak torque.

nodes) with a good accuracy is about 69.57 s by the single-core computing.

The tangential mesh effects on the accuracy and simulation speed are displaced in Fig. 31. And the computing time also increases linearly with the increment of number of mesh nodes. However, by comparing Figs. 30(a) and 31(a), it can be noticed that the 80000-node mesh (800×100 nodes, 205 s CPU time) in Fig. 30(a) needs more simulation time than the 120000-node mesh (300×400 nodes, 168 s CPU time) in Fig. 31(a). The reason is that the 80000-node mesh needs more nonlinear iterations for each time step, which further indicates that the solving time may be increased if there is a too big difference in the number of mesh nodes along radial and tangential directions. Furthermore, Fig. 31(b) and (c) also show that the accuracy on the average and peak-to-peak torque can be guaranteed if there are enough tangential nodes in the mesh.

Finally, the results show in Figs. 30 and 31 reveal that the proposed MRN has the potential to solve the large-scale model with hundreds of thousands of mesh nodes even though the single-core computing is used. A quicker simulation speed can be obtained if the number of mesh nodes along radial and tangential nodes are set close to each other. And the accuracy

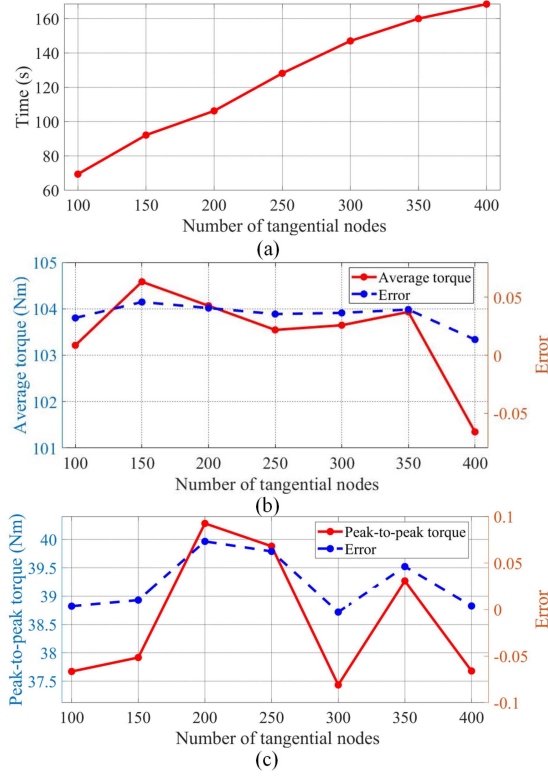


Fig. 31. Simulated torque and computing speed as function of number of tangential nodes by the proposed nodal-based MRN with 100 time steps. (the number of radial nodes is kept as a constant of 300, and the reference average torque and peak-to-peak torque are 100 Nm and 37.53 Nm solved by FEA). (a) Computing time by single core. (b) Average torque. (c) Peak-to-peak torque.

of the model will not be affected significantly if enough mesh nodes have already been utilized.

VI. CONCLUSION

The main contribution of this paper is to introduce a universal non-parametric mesh MRN model for radial interior PMSMs regarding pre-processing, solving, and post-processing aspects. The main benefits of the proposed methods can be concluded as follows

- Easy to implement.
- Automatic mesh and topology generation.
- Nonlinear material property and rotor movement can be considered.
- Both the periodic and anti-periodic boundary conditions can be modelled.
- Quick simulation speed and powerful parallel computing capability.
- High accuracy in both scalar and field quantities (e.g., torque waveform and field distribution).
- Fast convergence by the damping search algorithm.
- Capability of solving the large-scale model.

These features may make it possible to extend the usage of the universal and accurate MRN solver to the design and analysis of other electromagnetic devices. Meanwhile, the limitations of the MRN can also be noticed that sometimes a large-scaled mesh is

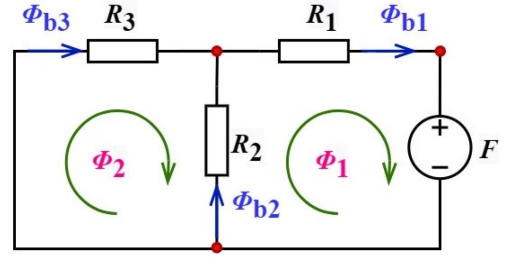


Fig. 32. An example of a simple MRN.

needed to get an acceptable accuracy, and the time step during time-stepping analysis is limited to a discrete variable and the possible time step is a multiple of a minimum time step. These limitations have already been mitigated a lot by introducing the damping coefficient search algorithm, proper initial condition and parallel computing, and they might be overcome by further improving the computing speed in the future e.g., by using adaptive mesh.

APPENDIX

Actually, the Jacobian matrix shown in [15] can be further transferred into the final form presented in (36). Fig. 32 shows a simple MRN, in which it contains one MMF source and three permeance, two nodes, three branches, and two loops in total.

When the loop analysis has been applied, the loop reluctance matrix $\mathbf{R}_l$ can be expressed

$$\mathbf{R}_{l,2 \times 2} = \mathbf{C}_3 \mathbf{R}_{b,3 \times 3} \mathbf{C}_{2 \times 3}^T \quad (37)$$

The branch reluctance $\mathbf{R}_b$ and loop matrix $\mathbf{C}$ can be written as

$$\mathbf{R}_{b,3 \times 3} = \begin{bmatrix} R_1 & & \\ & R_2 & \\ & & R_3 \end{bmatrix}, \mathbf{C}_3 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad (38)$$

where R_1 , R_2 and R_3 are the branch reluctance. The branch reluctance can be further given by

$$R = \frac{L_b}{\mu_0 \mu_r S_b} = \frac{L_b}{\mu_0 S_b} \frac{1}{\mu_r} = R_r \nu_r \quad (39)$$

where L_b and S_b are the length and area of the branch; ν_r is the relative nonlinear reluctivity; R_r is the reluctance without the consideration of the relative reluctivity part.

By substituting the matrices described in (37), (38), and (39) into (33), the residual function vector $\mathbf{f}_l$ is

$$\begin{aligned} \mathbf{f}_{l,2 \times 1} &= \mathbf{R}_{l,2 \times 2} \Phi_{2 \times 1} - \mathbf{F}_{1,2 \times 1} \\ &= \begin{bmatrix} R_{r1} \nu_{r1} + R_{r2} \nu_{r2} & -R_{r2} \nu_{r2} \\ -R_{r2} \nu_{r2} & R_{r2} \nu_{r2} + R_{r3} \nu_{r3} \end{bmatrix} \begin{bmatrix} \Phi_1 \\ \Phi_2 \end{bmatrix} - \begin{bmatrix} F \\ 0 \end{bmatrix} \end{aligned} \quad (40)$$

where Φ_1 and Φ_2 are the loop magnetic flux to be solved; F is the MMF shown in Fig. 32.

According to [15], the first entry of Jacobian matrix for loop analysis can be written as

$$\begin{aligned} J_1(1, 1) &= \frac{\partial f_1(1, 1)}{\partial \Phi_1} = R_{r1}\nu_{r1} + R_{r2}\nu_{r2} \\ &+ \left(R_{r1} \frac{\partial \nu_{r1}}{\partial \Phi_1} + R_{r2} \frac{\partial \nu_{r2}}{\partial \Phi_1} \right) \Phi_1 - R_{r2} \frac{\partial \nu_{r2}}{\partial \Phi_1} \Phi_2 \\ &= R_{r1} \left(\nu_{r1} + \frac{\partial \nu_{r1}}{\partial \Phi_1} \Phi_1 \right) + R_{r2} \left(\nu_{r2} + \frac{\partial \nu_{r2}}{\partial \Phi_1} (\Phi_1 - \Phi_2) \right) \end{aligned} \quad (41)$$

For the first partial derivative term in (41), using the chain rule

$$\frac{\partial \nu_{r1}}{\partial \Phi_1} \Phi_1 = \frac{\partial \nu_{r1}}{\partial B_1} \frac{\partial B_1}{\partial \Phi_1} \Phi_1 = \frac{\partial \nu_{r1}}{\partial B_1} \frac{1}{S_{b1}} \Phi_1 = \frac{\partial \nu_{r1}}{\partial B_1} B_1 \quad (42)$$

The relative reluctivity ν_r and differential relative reluctivity ν_{rd} can be defined from the nonlinear B - H curve as

$$\begin{cases} \nu_r = \frac{\mu_0 H}{B} \\ \nu_{rd} = \frac{\mu_0 dH}{dB} = \frac{d(\nu_r B)}{dB} = \nu_r + B \frac{d\nu_r}{dB} \end{cases} \quad (43)$$

By substituting the differential relative reluctivity ν_{rd} described in (43) into (42), (42) can be further expressed by the following form

$$\frac{\partial \nu_{r1}}{\partial \Phi_1} \Phi_1 = \frac{\partial \nu_{r1}}{\partial B_1} B_1 = \nu_{rd1} - \nu_{r1} \quad (44)$$

Hence, the first term in (41) can be written as

$$R_{r1} \left(\nu_{r1} + \frac{\partial \nu_{r1}}{\partial \Phi_1} \Phi_1 \right) = R_{r1} \nu_{rd1} = R'_1 \quad (45)$$

For the second partial derivative term in (41), using the chain rule

$$\begin{aligned} \frac{\partial \nu_{r2}}{\partial \Phi_1} (\Phi_1 - \Phi_2) &= \frac{\partial \nu_{r2}}{\partial B_2} \frac{\partial B_2}{\partial \Phi_1} (\Phi_1 - \Phi_2) = \frac{\partial \nu_{r2}}{\partial B_2} \frac{1}{S_{b2}} (\Phi_1 - \Phi_2) \\ &= \frac{\partial \nu_{r2}}{\partial B_2} \frac{1}{S_{b2}} \Phi_{b2} = \frac{\partial \nu_{r2}}{\partial B_2} B_2 \end{aligned} \quad (46)$$

Using the similar treatment as (44), (46) can be further written as

$$\frac{\partial \nu_{r2}}{\partial \Phi_1} (\Phi_1 - \Phi_2) = \frac{\partial \nu_{r2}}{\partial B_2} B_2 = \nu_{rd2} - \nu_{r2} \quad (47)$$

Therefore, the second term in (41) can be expressed as

$$R_{r2} \left(\nu_{r2} + \frac{\partial \nu_{r2}}{\partial \Phi_1} (\Phi_1 - \Phi_2) \right) = R_{r2} \nu_{rd2} = R'_2 \quad (48)$$

By substituting (45) and (48) into (41), the first entry of Jacobian matrix can be expressed with the differential reluctance as

$$J_1(1, 1) = R_{r1}\nu_{rd1} + R_{r2}\nu_{rd2} = R'_1 + R'_2 \quad (49)$$

Similarly, remaining elements of the Jacobian matrix can be derived by the same routine, and the complete entries of the Jacobian matrix can be written as

$$\mathbf{J}_1 = \begin{bmatrix} R_{r1}\nu_{rd1} + R_{r2}\nu_{rd2} & -R_{r2}\nu_{rd2} \\ -R_{r2}\nu_{rd2} & R_{r2}\nu_{rd2} + R_{r3}\nu_{rd3} \end{bmatrix}$$

$$= \begin{bmatrix} R'_1 + R'_2 & -R'_2 \\ -R'_2 & R'_2 + R'_3 \end{bmatrix} \quad (50)$$

Finally, the relatively complex Jacobian matrix presented in [15] has been fully transferred into the differential reluctance form as rewritten in (36) and (50), which finally gives the same result as shown in [32] and [50]. Besides, the same way can also be applied to derive the Jacobian matrix for the nodal analysis as expressed in (32).

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